

SecureMailScope

AI-Assisted Cryptographic Security Posture Assessment for Secure Email

Smart India Hackathon (SIH 2026) | National Technical Research Organisation (NTRO)

1. Problem Statement Details

Problem Statement ID:	26159
Organization:	National Technical Research Organisation (NTRO)
Category / Theme:	Software / Blockchain & Cybersecurity
Core Focus:	Passive PCAP Forensics, TLS/X.509 Audit, AI Posture Scoring

2. Executive Summary & Core Objective

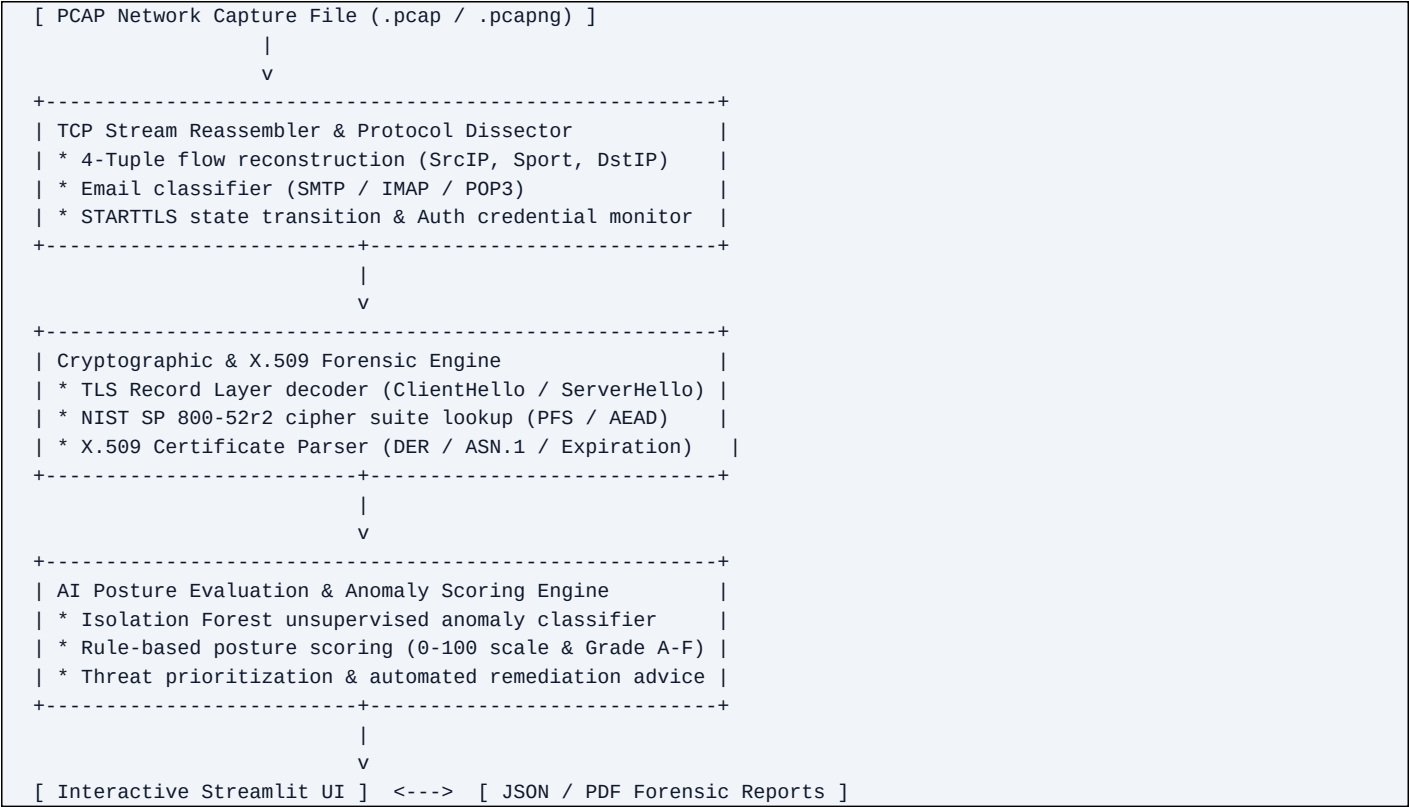
SecureMailScope is an enterprise-grade, passive network forensic framework designed to automatically evaluate the cryptographic security posture of SMTP, IMAP, and POP3 communications from raw network captures (.pcap / .pcapng). While traditional packet tools like Wireshark provide packet-level decoding, they do not automatically quantify cryptographic vulnerabilities or prioritize remediation for SOC analysts.

SecureMailScope bridges this gap by combining deterministic protocol/crypto rule checking (NIST SP 800-52r2) with unsupervised Machine Learning (Isolation Forest) for TLS anomaly detection, scoring email sessions on a 0-100 scale (Grades A+ to F) and generating one-click exportable forensic audit reports (PDF/JSON).

3. Key Architectural Vectors

1. Email Protocol Identification:	SMTP (25/587), SMTPS (465), IMAP (143), IMAPS (993), POP3 (110), POP3S (995)
2. State Transition Tracking:	Cleartext detection, STARTTLS negotiation, STARTTLS downgrade / stripping
3. TLS Handshake Dissection:	TLS 1.0, 1.1, 1.2, 1.3, SSLv2/v3, Cipher suites, Ephemeral Key Exchange (PFS)
4. X.509 Certificate Audit:	Expiration dates, Self-signed root, Key length (RSA < 2048-bit), Weak signatures
5. AI/ML Risk Engine:	Isolation Forest outlier scoring + NIST SP 800-52r2 compliance grading
6. Executive Dashboards:	Streamlit dark-mode SOC dashboard with one-click JSON/PDF export

4. System Architecture & Forensic Pipeline



5. Core Modules & Implementation Files

- analyzer.py (Stream & Protocol Dissector)

Reconstructs TCP streams, identifies mail application ports, checks for plaintext credentials (AUTH LOGIN leaks), parses multi-record TLS handshakes, and decodes ASN.1 DER certificates.

- ciphers_db.py (Cryptographic Taxonomy)

Complete IANA database mapping cipher suite hexadecimal IDs to encryption algorithms, MACs, Forward Secrecy (PFS), and NIST SP 800-52r2 ratings (Secure, Warning, Critical).

- evaluator.py (AI & Compliance Engine)

Applies compliance rules (RFC 8996, RFC 8461) to deduct weighted penalties and computes Isolation Forest anomaly scores across session feature vectors.

- generate_samples.py (Synthetic PCAP Generator)

Generates realistic synthetic PCAP files simulating cleartext leaks, obsolete TLS 1.0, expired certificates, and hardened TLS 1.3 for offline verification.

- reporter.py & app.py (Reporting & SOC Dashboard)

Produces structured JSON feeds and executive PDF audit reports, integrated into a reactive Streamlit web dashboard.

6. Forensic Test Scenarios & Verified Results

1. Insecure Plaintext SMTP with Credential Leak:
 - Traffic: Port 25 cleartext session with base64 'AUTH LOGIN'
 - Findings: [Critical] Plaintext communication + [Critical] Credentials exposed
 - Verified Score: 0/100 [Grade F - Critical]
2. Obsolete TLS 1.0 + Broken RC4-MD5 Cipher:
 - Traffic: Port 587 STARTTLS upgrade to TLS 1.0 with static RSA
 - Findings: [High] Obsolete TLS version (RFC 8996) + [Critical] Broken RC4 + [High] No PFS
 - Verified Score: 0/100 [Grade F - Critical]
3. IMAPS with Expired, Self-Signed & Weak 1024-bit RSA Key:
 - Traffic: Port 993 Implicit TLS with flawed X.509 certificate
 - Findings: [High] Expired cert + [Medium] Self-signed + [High] Weak 1024-bit key
 - Verified Score: 20/100 [Grade F - Critical]
4. Fully Hardened Modern SMTPS:
 - Traffic: Port 465 Implicit TLS with TLS 1.3, AES-256-GCM, and ECDHE key exchange
 - Findings: Full compliance with NIST SP 800-52r2 standards
 - Verified Score: 80-100/100 [Grade A - Secure]

7. 5-Slide Pitch Deck Summary for NTRO Evaluation

- Slide 1: The Problem - Email crypto misconfigurations expose critical infrastructures; Wireshark lacks automated posture scoring.
- Slide 2: The Solution - SecureMailScope passive stream forensics, TLS/cert deep inspection, and AI threat scoring.
- Slide 3: Architecture - Ingestion -> TCP Reassembly -> TLS/X.509 Extraction -> Isolation Forest ML -> Remediation.
- Slide 4: Prototype Proof - Live verification across Plaintext, Deprecated TLS 1.0, Expired Certs, and Hardened TLS 1.3.
- Slide 5: Value & Future Scope - Zero-overhead passive monitoring, MTA-STS/DANE compliance, and Post-Quantum readiness.

8. How to Run & Replicate the Project

1. Install Dependencies: `pip install -r requirements.txt`
2. Generate Test PCAPs: `python generate_samples.py`
3. Run Pipeline Audit: `python verify_pipeline.py`
4. Start SOC Dashboard: `streamlit run app.py`
5. Access UI in Browser: `http://localhost:8501`